

Objective

In this assignment you will use a unsupervised image segmentation algorithm to perform change detection (in the hypothesis the camera is still) between adjacent frames of a sequence.

Sketch of methodology

In particular, you will first use the SLIC algorithm to segment image frames and then classify a superpixel deciding whether it has changed or not from t to $t+1$, by comparing superpixels in a "similar position" at time t and $t+1$.

More specifically, given frames t and $t+1$ you should associate the superpixels at time t to those at time $t+1$, this should be done by designing a matching procedure between superpixels, inspired by the ones you saw in the previous lessons and by the fact the two frames should be similar enough.

Note that there isn't necessarily a superpixel at time $t+1$ to be associated to the previous frame (as there are new elements in the image). *Deal with this problem only if you have time.*

More details

You should identify a selection of appropriate data to present your results.

Provide visualization of images, superpixels and a mask of the detected changes at the superpixel level.

Use the code provided in the previous labs and implement new functions when needed. Add comments on your choices, for example on the feature(s) used for the matching procedure and the criterion used to detect the change.

Analyze the effect of choosing different number of superpixels.

The results don't need to be perfect, but any flaw or limit should be commented.

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